## EXERCISE 1

[6 marks]

Find and classify all points of inflection of $f(x) = 2 - x^3$

## EXERCISE 2

For $f(x) = x^3 - 3x^2 - 24x + 1$

**1.** Find the y-intercept of the function. [1 mark]

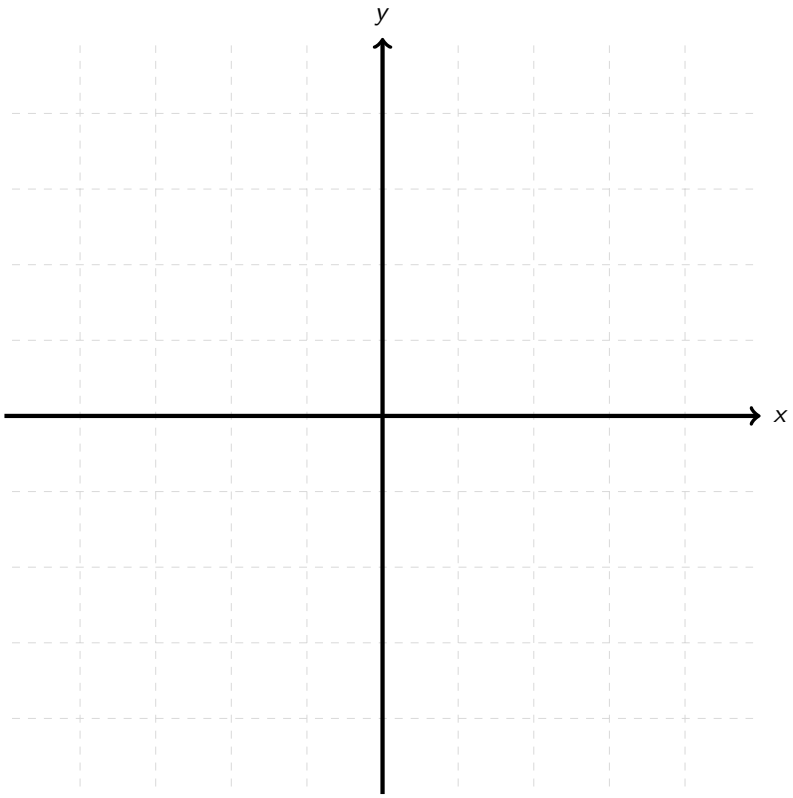
**2.** Find and classify all points where $f'(x) = 0$ [4 marks]

**3.** Find and classify all points of inflection. [3 marks]

**4.** Find intervals where the function is increasing or decreasing. [3 marks]

**5.** Find intervals where the function is concave up or down. [3 marks]

**6.** Sketch the function showing the features you have found. [3 marks]




**EXERCISE 3**[ONLY HL Students] [4 marks]

Prove that

$$\frac{\ln x}{x} \leq \frac{1}{e} \quad \text{for all } x > 0.$$

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